

SENTINEL

Technical Workflow & Integration Diagram Specification

Gujarat CCTV Integration Hackathon 2026 | Category 1 (Academic / Research / Startup)

Document: Workflow & Integration Diagram Specification

Project: Sentinel: Statewide CCTV Integration & ANPR Surveillance Engine

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1. Executive Summary

Sentinel is designed as a modular, edge-first statewide CCTV integration platform that ingests camera streams from heterogeneous departmental network feeds, performs continuous local license plate recognition (ANPR), correlates detections against real-time enforcement watchlists, dispatches low-latency WebSocket notifications, and reconstructs vehicle travel corridors on GIS maps with comprehensive audit logging.

This document details the operational workflow, end-to-end data pipelines, microservice integration points, database schemas, and edge-to-cloud scalability topology.

2. End-to-End Operational Workflow Diagram

The sequence below illustrates how a live RTSP feed is ingested, processed, matched against enforcement watchlists, and surfaced to operators:

[CAMERA ESTATE]

30 Sandbox RTSP Feeds (H.264/H.265)

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▼ (RTSP over TCP)

[1. CAPTURE & QUALITY GATE]

└─ OpenCV FFmpeg StreamCapture

└─ Presentation Timestamp (PTS) Sync

└─ Settling Window Gating (IDR Frames)

└─ Corrupt / Low-Luminance Frame Rejection

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▼ (Raw Video Frame 1920x1080)

[2. TILED INFERENCE & ANPR PIPELINE]

└─ Tiled Upscaling (2x3 Overlapping Grid @ 2x)

└─ YOLOv9-t End-to-End Plate Detector

└─ Bounding Box Remapping to Full Frame

└─ Fast-ALPR CCT-S-v2 Global OCR (10 Character Slots)

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▼ (Per-Frame Plate Candidates)

[3. NOISE REJECTION & TRACK VOTING]

└─ Camera Overlay Text Rejection (Timestamps, PTZ Labels, Billboards)

└─ IoU Spatial-Temporal Track Association (20-60 Frames per Vehicle)

└─ Character-Level Confidence-Weighted Voting (Right-Aligned Serial)

└─ Indian Plate Grammar Correction (O↔0, I↔1, S↔5, B↔8, Z↔2, G↔6)

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▼ (Validated Voted Plate Detection)

[4. FASTAPI INGEST & DB COMMIT]

└─ REST POST /api/v1/detections (Camera ID, Plate Text, Crop URI, PTS)

└─ PostgreSQL 17 + PostGIS + pg_trgm Transaction Commit

└─ Evidence Crop Saved to Local Storage (`/static/crops/`)

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▼ (Synchronous Pre-Commit Trigger)

[5. WATCHLIST CORRELATION & ALERTING]

└─ Exact String Match Check (Score: 1.0)

└─ Trigram Fuzzy Match Check (`pg\_trgm` Similarity > 0.7)

└─ Alert Row Created (`status = new`, Severity: Critical/High/Med)

└─ WebSocket Manager Fan-out (`ws://.../ws/alerts`)

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▼ (Live Alert Notification)

[6. OPERATOR UI & WORKFLOW ACTION]

└─ React + Leaflet Live Map Toast & Audio Alert

└─ Video Wall Tile Highlight (HLS Grid + Hero WebRTC)

└─ Operator Acknowledgment (`status = ack`, Stated Purpose)

└─ VAHAN Registry Enrichment (Owner, Make, Model, Fuel Type)

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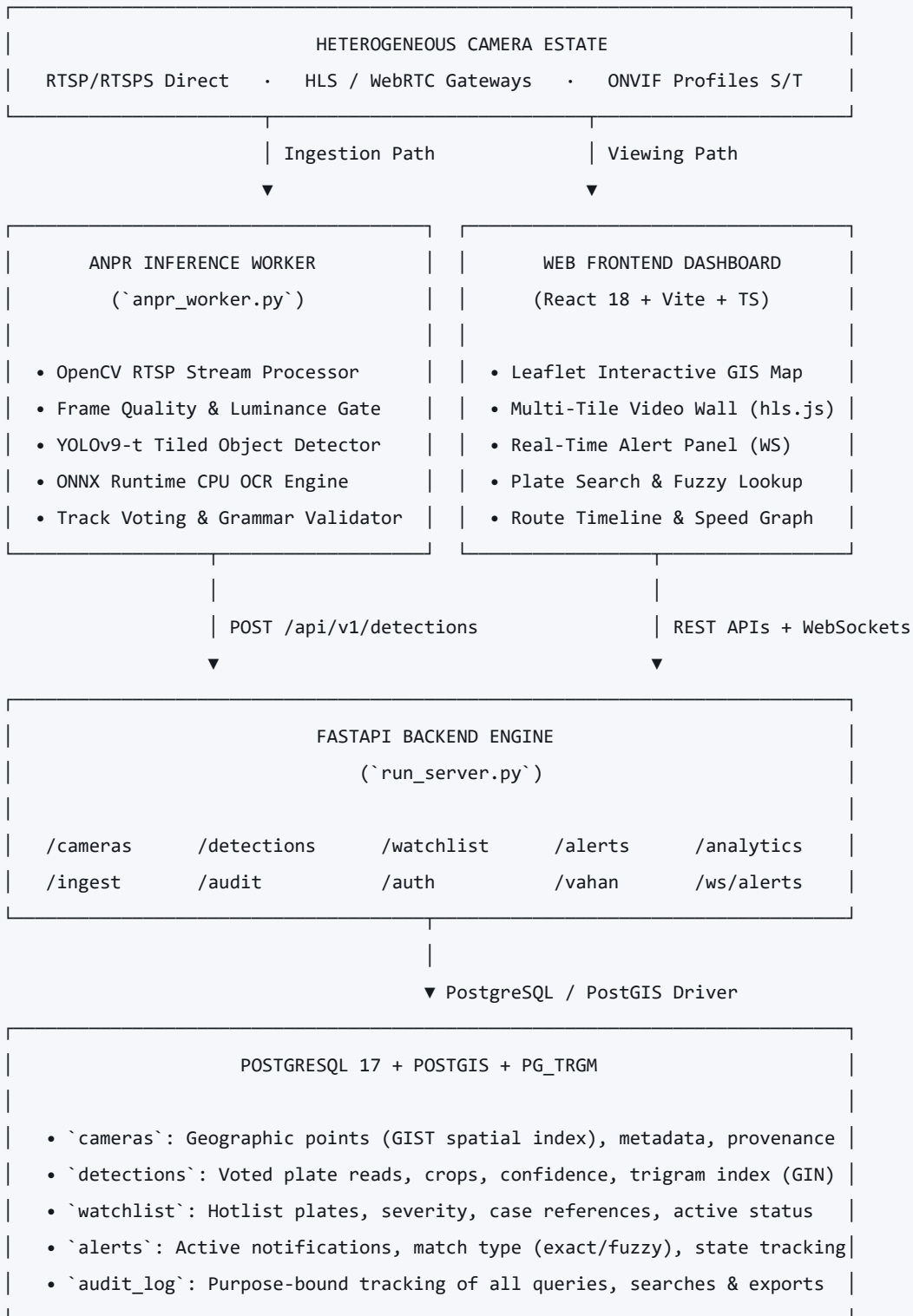
▼ (Investigation & Reporting)

[7. ROUTE RECONSTRUCTION & AUDIT]

- └ Chronological Sighting Query & GIS Point Ordering
- └ Great-Circle Distance, Time & Implied Speed Calculation
- └ Speed Violation Flagging (>150 km/h Impossible Transition Flag)
- └ Purpose-Bound Audit Logging (`audit\_log` Table)
- └ Excel (.xlsx) & PDF Output Evidence Report Export

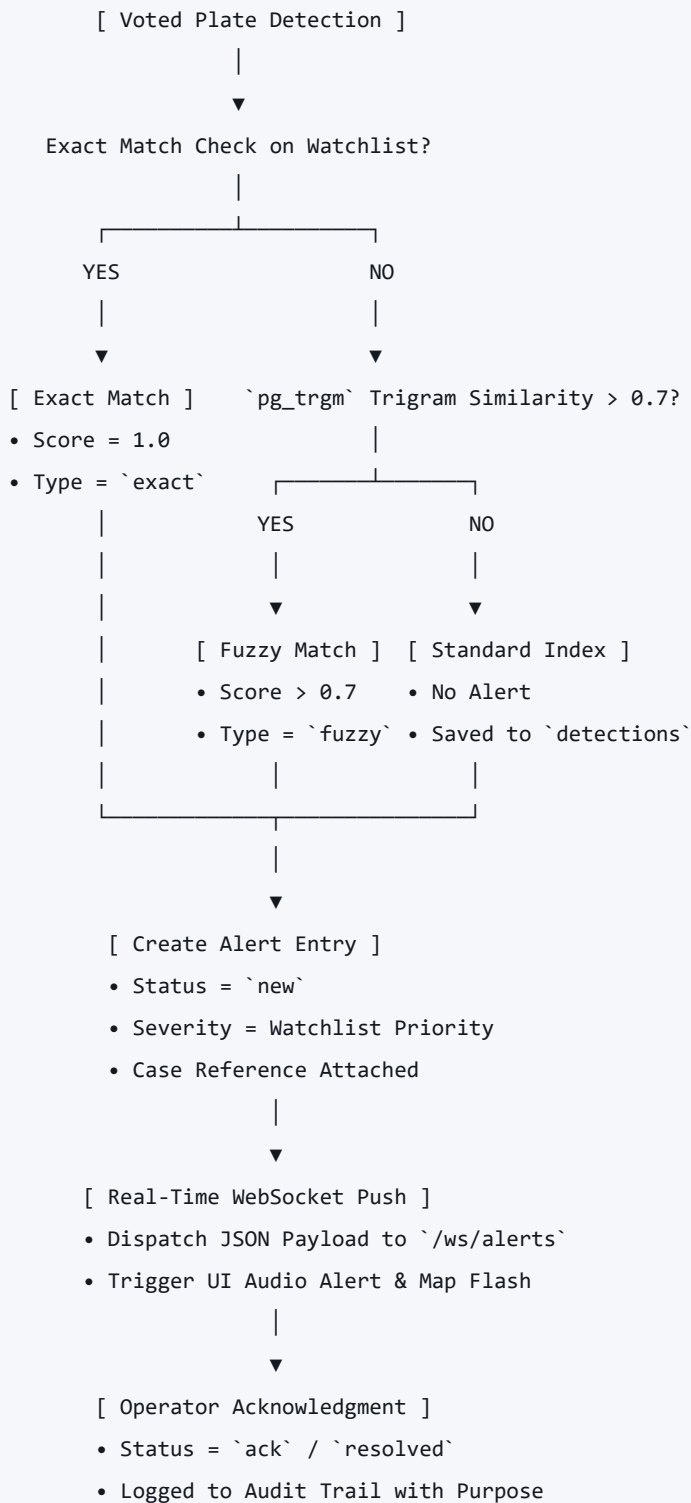
3. Microservice & System Integration Topology

Sentinel separates video ingestion and compute-heavy inference from the API web backend and frontend UI layer:



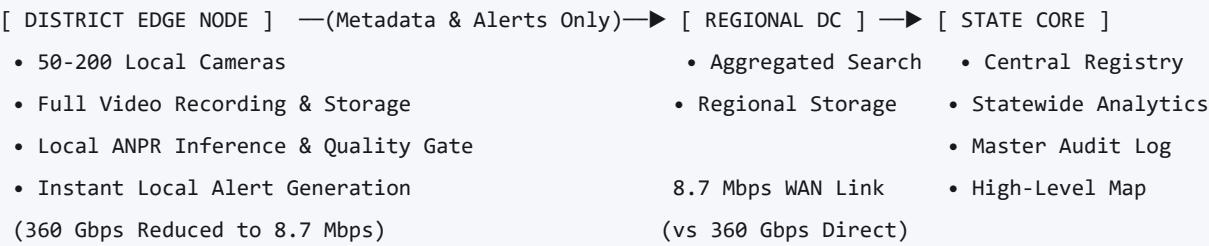
4. Watchlist Correlation & Alert Workflow

The diagram below details the exact decision tree executed whenever a number plate is indexed:



5. Statewide Edge-First Deployment Topology (Scale to 80,000 Cameras)

To prevent centralized network backhaul saturation (192 Gbps raw video bottleneck), Sentinel utilizes an **edge-first architecture**:



6. Integration Contract Interfaces

6.1 VAHAN Vehicle Registry Adapter Interface

- **Endpoint:** `GET /api/v1/vahan/vehicle/{plate_number}`
- **Mock / Integration Adapter:** Contract-first architecture returning registration date, owner name, vehicle class, fuel type, engine number, and insurance validity.

6.2 Audit Trail Verification Interface

- **Endpoint:** `GET /api/v1/analytics/audit`
- **Enforcement:** Logs user ID, role, client IP, action type (`SEARCH` , `ROUTE_RECONSTRUCT` , `EXPORT`), timestamp, and explicit **stated legal purpose**.

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